Regression Analysis — Solutions

Graduate entrance practice problems

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A. Simple linear regression

Solution 1 *Least squares in simple regression*

(a) The normal equations $\sum(y_i - \beta_0 - \beta_1 x_i) = 0$ and $\sum x_i(y_i - \beta_0 - \beta_1 x_i) = 0$ give $\beta_0 = \bar{y} - \beta_1 \bar{x}$ and then $\beta_1 S_{xx} = S_{xy}$. The objective is a convex quadratic with Hessian $2\begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}$, which has determinant $4nS_{xx} > 0$, so the stationary point is the global minimum.

(b) The first two identities are the normal equations, and $\sum \hat{y}_i e_i = \hat{\beta}_0 \sum e_i + \hat{\beta}_1 \sum x_i e_i = 0$. In $\sum(y_i - \bar{y})^2 = \sum\{e_i + (\hat{y}_i - \bar{y})\}^2$ the cross term is $2\sum e_i \hat{y}_i - 2\bar{y}\sum e_i = 0$.

(c) $SSR = \sum \hat{\beta}_1^2 (x_i - \bar{x})^2 = S_{xy}^2 / S_{xx}$, so $R^2 = S_{xy}^2/(S_{xx}S_{yy}) = r_{xy}^2$. Also $b = S_{xy}/S_{yy}$, so $\hat{\beta}_1 b = S_{xy}^2/(S_{xx}S_{yy}) = r_{xy}^2$.

Solution 2 *Inference in simple regression*

(a) $\bar{x} = 3$, $\bar{y} = 4.2$, $S_{xx} = 10$, $S_{xy} = 8$, so $\hat{\beta}_1 = 0.8$ and $\hat{\beta}_0 = 1.8$. $SST = 8.8$, $SSR = 0.8^2 \cdot 10 = 6.4$, $SSE = 2.4$, $MSE = 0.8$, $R^2 = 0.727$.

(b) $\mathrm{se}(\hat{\beta}_1) = \sqrt{0.8/10} = 0.283$; the interval is $0.8 \pm 3.182 \cdot 0.283 = (-0.10, 1.70)$. $t_0 = 2.83 < 3.182$: do not reject $\beta_1 = 0$ (consistent with the interval containing 0).

(c) $\hat{y} = 6.6$, and the prediction interval is $6.6 \pm 3.182\sqrt{0.8(1 + \frac{1}{5} + \frac{9}{10})} = 6.6 \pm 4.12$.

(d) $\mathrm{Cov}(\hat{\beta}_0, \hat{\beta}_1) = \mathrm{Cov}(\bar{y} - \hat{\beta}_1\bar{x}, \hat{\beta}_1) = \mathrm{Cov}(\bar{y}, \hat{\beta}_1) - \bar{x}\,\mathrm{Var}\,\hat{\beta}_1 = 0 - \bar{x}\sigma^2/S_{xx}$, using $\mathrm{Cov}(\bar{y}, \sum k_i y_i) = \frac{\sigma^2}{n}\sum k_i = 0$. They are uncorrelated iff $\bar{x} = 0$, e.g. after centring x.

Solution 3 *Regression through the origin*

(a) Minimizing $\sum(y_i - \beta x_i)^2$ gives $\hat{\beta} = \sum x_i y_i / \sum x_i^2 \sim N(\beta, \sigma^2 / \sum x_i^2)$. $X = \mathbf{x}$ has rank 1, so $SSE/\sigma^2 \sim \chi^2(n-1)$ and $MSE = SSE/(n-1)$ is unbiased.

(b) The only normal equation is $\sum x_i(y_i - \hat{\beta} x_i) = 0$. Nothing forces $\sum e_i = 0$, since $\mathbf{1}$ is not in the column space; e.g. $x = (1, 2)$, $y = (1, 1)$ gives $\hat{\beta} = 0.6$ and $\mathbf{e} = (0.4, -0.2)$.

(c) Without an intercept, $\sum(y_i - \bar{y})^2 = SSE + \sum(\hat{y}_i - \bar{y})^2$ fails, and SSE can exceed $\sum(y_i - \bar{y})^2$ (a line through the origin can fit worse than the horizontal line $\bar{y}$). The valid decomposition is the uncorrected one, $\sum y_i^2 = \sum \hat{y}_i^2 + \sum e_i^2$ (since $\hat{\mathbf{y}} \perp \mathbf{e}$), with the uncentred $R^2 = \sum \hat{y}_i^2 / \sum y_i^2$.

Solution 4 *Gauss–Markov for the slope*

(a) $E \sum c_i y_i = \beta_0 \sum c_i + \beta_1 \sum c_i x_i = \beta_1$ for all β_0, β_1 forces $\sum c_i = 0$ and $\sum c_i x_i = 1$.

(b) Let $k_i = (x_i - \bar{x})/S_{xx}$ and $d_i = c_i - k_i$. Then $\sum d_i = 0$ and $\sum d_i x_i = 0$, so $\sum k_i d_i = \frac{1}{S_{xx}} \sum (x_i - \bar{x}) d_i = 0$ and $\sum c_i^2 = \sum k_i^2 + \sum d_i^2 \geq \sum k_i^2$, with equality iff $d = 0$.

(c) $\operatorname{Var} \hat{\beta}_1 = \sigma^2 / S_{xx}$, and $S_{xx} \leq n$ on $[-1, 1]$ with equality iff half the points are at -1 and half at $+1$. (This design cannot detect curvature.)

Solution 5 *Expected mean squares*

(a) $E(SSR) = S_{xx} E\hat{\beta}_1^2 = S_{xx}(\sigma^2/S_{xx} + \beta_1^2)$. $E(SST) = E \sum \{\beta_1(x_i - \bar{x}) + \epsilon_i - \bar{\epsilon}\}^2 = \beta_1^2 S_{xx} + (n-1)\sigma^2$; subtract.

(b) $F_0 = \hat{\beta}_1^2 S_{xx} / MSE = t_0^2$.

(c) $SSR/\sigma^2 = (\hat{\beta}_1 \sqrt{S_{xx}}/\sigma)^2$ with $\hat{\beta}_1 \sqrt{S_{xx}}/\sigma \sim N(\beta_1 \sqrt{S_{xx}}/\sigma, 1)$, so it is $\chi^2(1, \lambda)$ with $\lambda = \beta_1^2 S_{xx}/(2\sigma^2)$ (half-convention).

Solution 6 *Inference on correlation*

(a) $SSR = r^2 SST$ and $SSE = (1 - r^2) SST$, so $t_0^2 = SSR/(SSE/(n-2)) = (n-2) r^2/(1 - r^2)$.

(b) $t_0 = 0.6\sqrt{26}/0.8 = 3.82 > 2.056$: reject $\rho = 0$.

(c) $z = \tanh^{-1} 0.6 = 0.693$, half-width $1.96/\sqrt{25} = 0.392$; back-transforming $(0.301, 1.085)$ gives $(0.29, 0.80)$.

Solution 7 *Simple regression with unequal variances* **[Past exam 2025-10 #4]**

(a) OLS: $\hat{\beta}_1 = S_{xy}/S_{xx}$, $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$.

(b) The log-likelihood is $-\frac{n_1}{2} \log \sigma_1^2 - \frac{n_2}{2} \log \sigma_2^2 - \frac{SS_1(\beta)}{2\sigma_1^2} - \frac{SS_2(\beta)}{2\sigma_2^2}$, with SS_g the group residual sums of squares. The MLE solves jointly $\hat{\sigma}_g^2 = SS_g(\hat{\beta})/n_g$ and $\hat{\beta}$ = the WLS estimator with weights $1/\hat{\sigma}_g^2$, computed iteratively; there is no closed form in general.

(c) The LSE is OLS as in (a). The WLSE is $(X'WX)^{-1} X'W\mathbf{y}$ with $W = \operatorname{diag}(1/\sigma_g^2)$, in practice with σ_g^2 estimated by group residual mean squares from an OLS fit.

(d) The MLE is a fixed point of iterated WLS with the variances estimated by SS_g/n_g. A one-step WLSE using OLS-based variance estimates (or divisors $n_g - 2$) differs from the MLE in general; they coincide when the WLS is iterated to convergence with the ML variance estimates.

Solution 8 *Lack of fit and replication*

(a) $y_{jk} - \hat{y}_j = (y_{jk} - \bar{y}_j) + (\bar{y}_j - \hat{y}_j)$, and the cross term vanishes because $\sum_k (y_{jk} - \bar{y}_j) = 0$ within each group.

(b) $SS_{PE}/\sigma^2 \sim \chi^2(n-m)$ regardless of the mean function, and it depends only on within-group deviations, independent of the group means. If the line is correct, SS_{LOF} is the residual sum of squares of a weighted regression of the m group means, so $SS_{LOF}/\sigma^2 \sim \chi^2(m-2)$.

(c) Without replicates, $SS_{PE} = 0$ with 0 degrees of freedom: there is no model-free estimate of σ^2 to compare the lack of fit with.

B. Least squares geometry and the hat matrix

Solution 9 *The hat matrix*

(a) $H' = H$, $H^2 = H$, $\operatorname{tr} H = \operatorname{tr}\{(X'X)^{-1}X'X\} = p+1$; then $(I-H)^2 = I - H$ and $\operatorname{tr}(I-H) = n-p-1$.

(b) $\mathbf{e} = (I-H)(X\beta+\epsilon) = (I-H)\epsilon$, so $\operatorname{Var}\mathbf{e} = \sigma^2(I-H)$, and $\operatorname{Cov}(\hat{\beta}, \mathbf{e}) = (X'X)^{-1}X'\sigma^2(I-H) = 0$.

(c) $h_{ii} = \sum_j h_{ij}^2 \geq h_{ii}^2$ gives $h_{ii} \leq 1$. With an intercept, $H = J/n + \tilde{H}$ with $\tilde{H}$ the projection onto the centred regressors, so $h_{ii} \geq 1/n$. If $h_{ii} = 1$, then $\sum_{j \neq i} h_{ij}^2 = 0$, so $\hat{y}_i = y_i$ and $e_i = 0$ (also $\operatorname{Var} e_i = 0$).

(d) $\tilde{H} = \tilde{\mathbf{x}}\tilde{\mathbf{x}}'/S_{xx}$ with $\tilde{\mathbf{x}} = (x_i - \bar{x})_i$, so $h_{ii} = \frac{1}{n} + (x_i - \bar{x})^2/S_{xx}$.

Solution 10 *Quadratic forms*

(a) $E\mathbf{y}'A\mathbf{y} = E\operatorname{tr}(A\mathbf{y}\mathbf{y}') = \operatorname{tr}\{A(\Sigma + \mu\mu')\}$.

(b) With $\mu = X\beta$, $\Sigma = \sigma^2 I$: $SSE = \mathbf{y}'(I-H)\mathbf{y}$ gives $(n-p-1)\sigma^2$, since $(I-H)X = 0$. $SSR = \mathbf{y}'(H-J/n)\mathbf{y}$ gives $p\sigma^2 + \beta'X'(I-J/n)X\beta$, and $SST = \mathbf{y}'(I-J/n)\mathbf{y}$ gives $(n-1)\sigma^2 + \beta'X'(I-J/n)X\beta$.

(c) Both matrices are symmetric idempotent (using $HJ = J$) with $(H-J/n)(I-H) = 0$, so the forms are independent, $SSR/\sigma^2 \sim \chi^2(p, \lambda)$ and $SSE/\sigma^2 \sim \chi^2(n-p-1)$.

Solution 11 *Centred regression and SSR* **[Past exam 2021-04 #1]**

(a) $SSR = \mathbf{y}'X(X'X)^{-1}X'\mathbf{y} = \mathbf{y}'H\mathbf{y}$ and $SST = \mathbf{y}'\mathbf{y}$ (centred data).

(b) $\langle X\beta, \mathbf{y}\rangle = \langle X\beta, H\mathbf{y}\rangle$ because $X\beta$ lies in the column space. By Cauchy–Schwarz, $\langle X\beta, H\mathbf{y}\rangle^2 \leq \|X\beta\|^2\|H\mathbf{y}\|^2$, so the ratio is at most $\|H\mathbf{y}\|^2/\|\mathbf{y}\|^2 = SSR/SST$, with equality when $X\beta \propto H\mathbf{y}$.

(c) $\beta = c\hat{\beta}$ for any $c \neq 0$: the maximum squared correlation between $\mathbf{y}$ and a linear combination of the regressors is attained by the fitted values (the multiple correlation).

Solution 12 *SVD and eigendecomposition* **[Past exam 2022-04 #4]**

$X = UDV'$ with U ($n \times p$, orthonormal columns), $D = \mathrm{diag}(d_1 \geq \cdots \geq d_p \geq 0)$ the singular values, and V ($p \times p$ orthogonal). Then $X'X = VD^2V'$: the eigenvectors of $X'X$ are the right singular vectors and the eigenvalues are d_j^2. Also $XV = UD$: the principal components $X\mathbf{v}_j = d_j\mathbf{u}_j$.

Solution 13 *Partial regression*

(a) Multiply the normal equations $\begin{pmatrix} X_2'X_2 & X_2'X_1 \\ X_1'X_2 & X_1'X_1 \end{pmatrix} \binom{\hat\beta_2}{\hat\beta_1} = \binom{X_2'\mathbf{y}}{X_1'\mathbf{y}}$ by $\begin{pmatrix} I & -X_2'X_1(X_1'X_1)^{-1} \\ 0 & I \end{pmatrix}$. The first row becomes $X_2'(I - H_1)X_2\hat\beta_2 = X_2'(I - H_1)\mathbf{y}$.

(b) The full residual $\mathbf{e}$ is orthogonal to X_1, so $\mathbf{e} = (I - H_1)\mathbf{e} = (I - H_1)\mathbf{y} - (I - H_1)X_2\hat\beta_2$, which is the residual of the second regression by (a).

(c) Then $X'X$ is block diagonal, so $\hat\beta_1 = (X_1'X_1)^{-1}X_1'\mathbf{y}$ in both models, $H = H_1 + H_2$, and SS due to the regressors splits as $\mathbf{y}'H_1\mathbf{y} + \mathbf{y}'H_2\mathbf{y}$ (centring both blocks when there is an intercept).

Solution 14 *Frisch–Waugh–Lovell* **[Past exam 2022-04 #6]**

(i) Problem 13(a)–(b) (Frisch–Waugh–Lovell).

(ii) A multiple regression coefficient is the effect of x_2 on y after both have been purged of their linear association with x_1: the partial effect holding x_1 fixed.

Solution 15 *Projections and partial regression* **[Past exam 2020-10 #5]**

(a) $\hat\beta_2 = \{X_2'(I - H_1)X_2\}^{-1}X_2'(I - H_1)\mathbf{y}$ (Problem 13).

(b) If $X_1'X_2 = 0$, the column spaces are orthogonal and $H = H_1 + H_2$, so $H\mathbf{y} - H_1\mathbf{y} = H_2\mathbf{y} = X_2(X_2'X_2)^{-1}X_2'\mathbf{y}$.

Solution 16 *Full versus reduced model estimators* **[Past exam 2023-10 #4]**

With $A = (X_k'X_k)^{-1}X_k'X_r$ and $B = X_r'(I - H_k)X_r$, the partitioned inverse gives $\mathrm{Var}\,\hat\beta_k = \sigma^2\{(X_k'X_k)^{-1} + AB^{-1}A'\}$, while $\mathrm{Var}\,\tilde\beta_k = \sigma^2(X_k'X_k)^{-1}$. The difference $\sigma^2AB^{-1}A'$ is nonnegative definite.

Solution 17 *Properties of least squares fits* **[Past exam 2024-10 #4]**

(a) $\mathbf{y} = X\mathbf{e}_j$ lies in the column space, so $\hat\beta = \mathbf{e}_j$.

(b) As in Problem 19 (without centring): the maximum is $\|H\mathbf{y}\|^2/\|\mathbf{y}\|^2 = \sum \hat y_i^2 / \sum y_i^2$. In case (a) it equals 1.

Solution 18 *Residuals and fitted values* **[Past exam 2025-04 #4]**

(1) $E\mathbf{e}'\mathbf{e} = \sigma^2\,\mathrm{tr}(I - H) = (n - k - 1)\sigma^2$, and $\mathrm{Cov}(\hat\beta, \mathbf{e}) = (X'X)^{-1}X'\sigma^2(I - H) = 0$.

(2) $\mathrm{Var}\,\hat{\mathbf{y}} = \sigma^2 H$, so $\sum \mathrm{Var}\,\hat y_i = \sigma^2\,\mathrm{tr}\,H$ and $c = k + 1$.

(3) $HJ = JH = J$ (since $H\mathbf{1} = \mathbf{1}$), so $(H - J/n)^2 = H - 2J/n + J/n = H - J/n$.

C. Inference in the linear model

Solution 19 *Overall F test*

(a) Follows from Solution 10(c) and the definition of noncentral F.

(b) $2\sigma^2\lambda = \|(I - J/n)X\beta\|^2$, the squared length of the centred mean vector $\sum_{j\geq 1}\beta_j(\mathbf{x}_j - \bar{x}_j\mathbf{1})$. The centred columns are linearly independent (full rank), so this is 0 iff $\beta_1 = \cdots = \beta_p = 0$.

(c) $F_0 = \dfrac{R^2/p}{(1-R^2)/(n-p-1)}$, dividing SSR and SSE by SST.

Solution 20 R^2 *of nested models* **[Past exam 2025-04 #5]**

SST is the same for both models, and $SSE(F) \leq SSE(R)$ because the reduced model minimizes over a subspace. Hence $R^2(F) = 1 - SSE(F)/SST \geq 1 - SSE(R)/SST = R^2(R)$.

Solution 21 *Partial and sequential F*

(a) With $C = \mathbf{e}_j'$, Theorem (restricted LS) gives $SSE(R) - SSE = \hat{\beta}_j^2/c_{jj}$, and the partial F is $\hat{\beta}_j^2/(c_{jj}MSE) = t_0^2$.

(b) The sum telescopes: $\sum_{j=1}^{p}\{SSR(x_1,\ldots,x_j) - SSR(x_1,\ldots,x_{j-1})\} = SSR(x_1,\ldots,x_p) - 0$.

(c) With orthogonal centred regressors, $H - J/n = \sum_j \tilde{\mathbf{x}}_j\tilde{\mathbf{x}}_j'/\|\tilde{\mathbf{x}}_j\|^2$, so adding x_j always contributes $(\tilde{\mathbf{x}}_j'\mathbf{y})^2/\|\tilde{\mathbf{x}}_j\|^2$, regardless of what is already in the model.

Solution 22 *Restricted least squares*

(a) $\hat{\beta} = \frac{1}{200}(220\cdot 47 - 90\cdot 196 + 70\cdot 112,\ -90\cdot 47 + 65\cdot 196 - 75\cdot 112,\ 70\cdot 47 - 75\cdot 196 + 105\cdot 112)' = (2.7, 0.55, 1.75)'$. $\mathbf{y}'\mathbf{y} = 431$ and $\hat{\beta}'X'\mathbf{y} = 2.7\cdot 47 + 0.55\cdot 196 + 1.75\cdot 112 = 430.7$, so $SSE = 0.3$ and $MSE = 0.1$.

(b) $C = (0, 1, -1)$: $C\hat{\beta} = -1.2$ and $C(X'X)^{-1}C' = (65 + 105 + 150)/200 = 1.6$, so $Q = 1.44/1.6 = 0.9$ and $F_0 = 0.9/0.1 = 9.0 < 10.13$. Do not reject.

(c) Under H_0, $y = \beta_0 + \beta_1(x_1 + x_2) + \epsilon$. With $z = x_1 + x_2 = (1, 3, 4, 7, 7, 10)$ the fit is $\hat{y} = 2.1 + 1.075z$ with $SSE(R) = 1.2$. So $SSE(R) - SSE = 0.9 = Q$, as it must be.

Solution 23 *Constrained least squares* **[Past exam 2026-04 #4]**

(a) Lagrange: $X'X\beta + C\mu = X'\mathbf{y}$ and $C'\beta = \mathbf{0}$ give $\mu = \{C'(X'X)^{-1}C\}^{-1}C'\hat{\beta}$ and the stated $\hat{\beta}_r$.

(b) Fit the full and the restricted model. Then $F = \frac{\{SSE(R) - SSE(F)\}/q}{SSE(F)/(n-p-1)} \sim F(q, n-p-1)$ under H_0, because $SSE(R) - SSE(F) = \mathbf{y}'(H - H_R)\mathbf{y}$ with $H - H_R$ idempotent of rank q and orthogonal to $I - H$.

(c) $C'\hat{\beta} \sim N(C'\beta, \sigma^2 C'(X'X)^{-1}C)$, so $F = \frac{(C'\hat{\beta})'\{C'(X'X)^{-1}C\}^{-1}C'\hat{\beta}/q}{MSE} \sim F(q, n-p-1)$ under H_0.

(d) Yes: $SSE(R) - SSE(F) = (\hat{\beta} - \hat{\beta}_r)'X'X(\hat{\beta} - \hat{\beta}_r) = (C'\hat{\beta})'\{C'(X'X)^{-1}C\}^{-1}C'\hat{\beta}$, so the statistics are identical.

Solution 24 *Least squares under a linear constraint* **[Past exam 2023-10 #5]**

The full parameter space $\{\theta : \mathbf{1}'\theta = 0\}$ has dimension 3, so $SSE(F) = \|\mathbf{y} - \hat{\theta}\|^2$ is the squared projection of $\mathbf{y}$ on $\mathbf{1}$: $(\sum y_i)^2/4$, with 1 degree of freedom. Under H_0 the space is $\{\mathbf{1}'\theta = 0,\ \theta_1 = \theta_3\}$, whose orthogonal complement is spanned by the orthogonal vectors $\mathbf{1}$ and $(1, 0, -1, 0)'$. So $SSE(R) = (\sum y_i)^2/4 + (y_1 - y_3)^2/2$. Hence

$$F_0 = \frac{\{SSE(R) - SSE(F)\}/1}{SSE(F)/1} = \frac{2(y_1 - y_3)^2}{(y_1 + y_2 + y_3 + y_4)^2} \sim F(1, 1) \text{ under } H_0.$$

Solution 25 *Simultaneous inference*

(a) $\hat{\beta} - \beta = (X'X)^{-1}X'\epsilon$, so $(\hat{\beta} - \beta)'X'X(\hat{\beta} - \beta) = \epsilon' H \epsilon \sim \sigma^2\chi^2(p + 1)$, independent of $SSE = \epsilon'(I - H)\epsilon$.

(b) By Cauchy–Schwarz, $\max_{\mathbf{a}} \frac{\{\mathbf{a}'(\hat{\beta}-\beta)\}^2}{\mathbf{a}'(X'X)^{-1}\mathbf{a}} = (\hat{\beta} - \beta)'X'X(\hat{\beta} - \beta)$. So, with probability $1 - \alpha$, for all $\mathbf{a}$, $|\mathbf{a}'(\hat{\beta} - \beta)| \le \sqrt{(p + 1)F_\alpha(p + 1, n - p - 1)}\sqrt{\mathbf{a}'(X'X)^{-1}\mathbf{a}\, MSE}$.

(c) The multipliers are 2.086 (individual), 2.613 (Bonferroni) and $\sqrt{3 \cdot 3.098} = 3.049$ (Scheffé). Bonferroni is better for three prespecified coefficients; Scheffé covers all combinations.

Solution 26 *Comparing two regressions*

(a) $E(y) = \beta_{01} + \beta_{11}x + \gamma_0 d + \gamma_1 xd$ with $\gamma_0 = \beta_{02} - \beta_{01}$ and $\gamma_1 = \beta_{12} - \beta_{11}$.

(b) The design is a reparametrization of the block-diagonal design $\text{diag}\{(\mathbf{1}, \mathbf{x}_1), (\mathbf{1}, \mathbf{x}_2)\}$: both span the same column space. The fitted values, and hence SSE, are those of the separate fits.

(c) Identical lines: $H_0 : \gamma_0 = \gamma_1 = 0$, with $F = \frac{\{SSE(R)-SSE(F)\}/2}{SSE(F)/(n_1+n_2-4)} \sim F(2, n_1 + n_2 - 4)$. Parallel lines: $H_0 : \gamma_1 = 0$, with a t test on $n_1 + n_2 - 4$ degrees of freedom (equivalently $F(1, n_1 + n_2 - 4)$).

Solution 27 *Indicator variables*

(a) $d_1 + d_2 + d_3 = \mathbf{1}$, the intercept column, so the columns are dependent and $X'X$ is singular.

(b) β_1 is the common slope; β_0 the intercept of the baseline level; β_2, β_3 the differences between levels 1, 2 and the baseline at any fixed x.

(c) $H_0 : \beta_2 = \beta_3 = 0$ by a partial F test with $(2, n - 4)$ degrees of freedom.

Solution 28 *Piecewise linear regression*

(a) For $x \le c$, $E(y) = \beta_0 + \beta_1 x$; for $x > c$, $(\beta_0 - \beta_2 c) + (\beta_1 + \beta_2)x$. Both equal $\beta_0 + \beta_1 c$ at c. The slopes are β_1 and $\beta_1 + \beta_2$.

(b) Add $\beta_3 \mathbf{1}\{x > c\}$.

(c) t test of $H_0 : \beta_2 = 0$.

Solution 29 *Two-way ANOVA without replication* **[Past exam 2024-05 #5]**

(1) Q_1 is a function of the row means $\bar{y}_{i\cdot}$, and Q_2 is a function of the interaction residuals $r_{ij} = y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}$. These are jointly normal and uncorrelated: $\mathrm{Cov}(\bar{y}_{i\cdot}, r_{kl}) = \frac{\sigma^2}{b}\{\delta_{ik} - \frac{1}{a} - \delta_{ik} + \frac{1}{a}\} = 0$ (row averaging kills the column-centred residual). Hence they are independent. Equivalently, the projection matrices P_A and P_{AB} satisfy $P_A P_{AB} = 0$.

(2) $Q_2/\sigma^2 \sim \chi^2((a-1)(b-1))$, so $\hat{\sigma}^2 = Q_2/\{(a-1)(b-1)\}$. The estimator $\bar{y}_{1\cdot} + \bar{y}_{\cdot 1} - \bar{y}_{\cdot\cdot}$ of $\mu + \alpha_1 + \beta_1$ has variance

$$\sigma^2\Big(\frac{1}{b} + \frac{1}{a} - \frac{1}{ab}\Big) = \sigma^2 \frac{a+b-1}{ab},$$

so the interval is $\bar{y}_{1\cdot} + \bar{y}_{\cdot 1} - \bar{y}_{\cdot\cdot} \pm t_{\alpha/2}((a-1)(b-1))\sqrt{\hat{\sigma}^2(a+b-1)/(ab)}$.

D. Diagnostics and model selection

Solution 30 *Deletion formulas*

(a) $X'_{(i)}X_{(i)} = A - \mathbf{x}_i\mathbf{x}_i'$ and $X'_{(i)}\mathbf{y}_{(i)} = X'\mathbf{y} - \mathbf{x}_i y_i$, with $A = X'X$. Then

$$\hat{\beta}_{(i)} = \Big(A^{-1} + \frac{A^{-1}\mathbf{x}_i\mathbf{x}_i'A^{-1}}{1-h_{ii}}\Big)(X'\mathbf{y} - \mathbf{x}_i y_i) = \hat{\beta} - A^{-1}\mathbf{x}_i y_i + \frac{A^{-1}\mathbf{x}_i(\hat{y}_i - h_{ii}y_i)}{1-h_{ii}} = \hat{\beta} - \frac{A^{-1}\mathbf{x}_i e_i}{1-h_{ii}}.$$

(b) For $j \neq i$, the new residual is $e_j + h_{ij}e_i/(1-h_{ii})$. Use $\sum_{j\neq i} h_{ij}e_j = -h_{ii}e_i$ (because $H\mathbf{e} = \mathbf{0}$) and $\sum_{j\neq i} h_{ij}^2 = h_{ii} - h_{ii}^2$:

$$SSE_{(i)} = (SSE - e_i^2) - \frac{2h_{ii}e_i^2}{1-h_{ii}} + \frac{h_{ii}e_i^2}{1-h_{ii}} = SSE - \frac{e_i^2}{1-h_{ii}}.$$

(c) $D_i = \frac{(\hat{\beta}-\hat{\beta}_{(i)})'A(\hat{\beta}-\hat{\beta}_{(i)})}{(p+1)s^2} = \frac{h_{ii}e_i^2}{(1-h_{ii})^2(p+1)s^2} = \frac{r_i^2}{p+1}\cdot\frac{h_{ii}}{1-h_{ii}}$.

Solution 31 *Studentized residuals and PRESS*

(a) $y_i - \mathbf{x}_i'\hat{\beta}_{(i)} = e_i/(1-h_{ii})$ is $N(0, \sigma^2/(1-h_{ii}))$, because $1 + \mathbf{x}_i'(X'_{(i)}X_{(i)})^{-1}\mathbf{x}_i = 1/(1-h_{ii})$. It is independent of $s^2_{(i)}$, which comes from the other observations, with $(n-p-2)s^2_{(i)}/\sigma^2 \sim \chi^2(n-p-2)$.

(b) Substitute $s^2_{(i)} = s^2(n-p-1-r_i^2)/(n-p-2)$ from Solution 17(b). Since $s^2_{(i)} \geq 0$, $r_i^2 \leq n-p-1$: the internally studentized residual is bounded, another sign that it is not t-distributed.

(c) The i-th leave-one-out prediction error is $y_i - \mathbf{x}_i'\hat{\beta}_{(i)} = e_i/(1-h_{ii})$ by Solution 17(a).

Solution 32 *Studentized residuals* **[Past exam 2026-04 #5]**

(a) $\mathbf{e} = (I-H)\mathbf{y} \sim N(\mathbf{0}, \sigma^2(I-H))$, so $e_i \sim N(0, \sigma^2(1-h_{ii}))$.

(b) e_i and s are dependent (e_i^2 is part of SSE), so the ratio is not a normal over an independent chi; indeed $r_i^2 \leq n-p-1$ is bounded.

(c) Problem 30(a)–(b).

Solution 33 *Leverage, outliers and influence* **[Past exam 2022-04 #5]**

Leverage: $h_{ii} = \mathbf{x}_i'(X'X)^{-1}\mathbf{x}_i$, equivalently the Mahalanobis distance of $\mathbf{x}_i$; flag $h_{ii} > 2(p+1)/n$. *Outliers*: studentized residuals $r_i^* = e_i/(s_{(i)}\sqrt{1-h_{ii}}) \sim t(n-p-2)$; flag large $|r_i^*|$, with a Bonferroni correction. *Influence*: Cook's distance $D_i = \frac{r_i^2}{p+1}\frac{h_{ii}}{1-h_{ii}}$ (compare with $F_{0.5}(p+1, n-p-1)$), DFFITS_i (flag $> 2\sqrt{(p+1)/n}$), DFBETAS and COVRATIO. Then inspect flagged cases for errors, and refit with and without them.

Solution 34 *Selection criteria*

(a) Adding a regressor enlarges the column space, so the minimum SSE cannot increase.

(b) R_a^2 increases iff MSE decreases: $\frac{SSE_{k+1}}{n-k-2} < \frac{SSE_k}{n-k-1}$. With $F = \frac{SSE_k - SSE_{k+1}}{SSE_{k+1}/(n-k-2)}$ this is equivalent to $F > 1$.

(c) If the model is correct, $E(SSE_k) = (n-k-1)\sigma^2$, and with $\hat\sigma^2 \approx \sigma^2$, $E(C_k) \approx (n-k-1) + 2(k+1) - n = k+1$. For the full model $SSE_p/\hat\sigma^2 = n-p-1$ exactly, so $C_p = p+1$.

Solution 35 *Mallows' C_p* **[Past exam 2025-04 #6]**

If $\mu = X_p\beta_p$, then $(I-H_p)\mu = \mathbf{0}$ and $E\,SSE(p) = \sigma^2\,\mathrm{tr}(I-H_p) = (n-p)\sigma^2$. With $\hat\sigma^2$ (from the full model) approximately σ^2, $E(C_p) \approx (n-p) - n + 2p = p$.

Solution 36 *Out-of-sample prediction error* **[Past exam 2021-10 #1]**

(1) Let X_1 be the training design. For $i > n_1$, y_i is independent of $\hat\beta$, and $y_i - \mathbf{x}_i'\hat\beta$ has mean 0 and variance $\sigma^2\{1 + \mathbf{x}_i'(X_1'X_1)^{-1}\mathbf{x}_i\}$. So the expectation is

$$\sigma^2 + \frac{\sigma^2}{n-n_1}\sum_{i>n_1}\mathbf{x}_i'(X_1'X_1)^{-1}\mathbf{x}_i.$$

(2) $E\frac{1}{n_1}\sum_{i\le n_1}(y_i - \mathbf{x}_i'\hat\beta)^2 = \frac{n_1-p}{n_1}\sigma^2 < \sigma^2$, which is below (1). So the training error is biased downward and is not unbiased; it underestimates the prediction error.

E. Departures from assumptions and remedies

Solution 37 *Transformations*

(a) $y^\lambda = e^{\lambda\log y} = 1 + \lambda\log y + O(\lambda^2)$.

(b) $\mathrm{Var}\,h(y) \approx h'(\mu)^2 g(\mu)$, so take $h'(\mu) \propto g(\mu)^{-1/2}$: $g = \mu$ gives $h = \sqrt{y}$, and $g = \mu^2$ gives $h = \log y$.

(c) $y^{(\lambda)}$ has different units for different λ, so its SSE values are not comparable. The density of y includes the Jacobian $\prod y_i^{\lambda-1} = \dot y^{n(\lambda-1)}$; dividing by $\dot y^{\lambda-1}$ absorbs it, and then the profile log-likelihood is $-\frac{n}{2}\log SSE(\lambda)$+const. So minimizing SSE of $z^{(\lambda)}$ is maximum likelihood.

Solution 38 *Errors in variables*

(a) $\hat\beta_1 \xrightarrow{p} \operatorname{Cov}(x,y)/\operatorname{Var} x = \beta_1\sigma_*^2/(\sigma_*^2+\sigma_\delta^2)$: attenuation towards 0.

(b) $\hat\beta_1/\hat\kappa$ with the reliability $\hat\kappa = (s_x^2-\sigma_\delta^2)/s_x^2$, which converges to $\sigma_*^2/(\sigma_*^2+\sigma_\delta^2)$; equivalently $S_{xy}/(S_{xx}-(n-1)\sigma_\delta^2)$.

(c) Yes: $y = \beta_0+\beta_1 x+(\epsilon+\beta_1\delta)$, and the new error has mean 0 and is independent of the fixed x; only the error variance increases.

Solution 39 *Omitted variables*

(a) $E\tilde\beta_1 = \frac{\sum(x_{i1}-\bar x_1)Ey_i}{S_{11}} = \beta_1+\beta_2\frac{S_{12}}{S_{11}}$.

(b) $\operatorname{Var}\tilde\beta_1 = \sigma^2/S_{11}$, while $\operatorname{Var}\hat\beta_1 = \sigma^2/\{S_{11}(1-r_{12}^2)\}$. So dropping x_2 reduces the variance by the factor $1-r_{12}^2$.

(c) $\operatorname{MSE}(\tilde\beta_1) = \frac{\sigma^2}{S_{11}}+\beta_2^2\frac{S_{12}^2}{S_{11}^2}$ and $\operatorname{Var}\hat\beta_1 - \frac{\sigma^2}{S_{11}} = \frac{\sigma^2 r_{12}^2}{S_{11}(1-r_{12}^2)}$. Since $\operatorname{Var}\hat\beta_2 = \sigma^2/\{S_{22}(1-r_{12}^2)\}$ and $r_{12}^2 = S_{12}^2/(S_{11}S_{22})$, the inequality $\operatorname{MSE}(\tilde\beta_1)\le\operatorname{Var}\hat\beta_1$ reduces to $\beta_2^2\le\operatorname{Var}\hat\beta_2$.

Solution 40 *Weighted least squares*

(a) $V=\operatorname{diag}(x_i)$, so GLS minimizes $\sum(y_i-\beta x_i)^2/x_i$: $\hat\beta_G = \sum y_i/\sum x_i$ (the ratio estimator), with $\operatorname{Var}\hat\beta_G = \sigma^2/\sum x_i$.

(b) $\operatorname{Var}\hat\beta_{OLS} = \sigma^2\sum x_i^3/(\sum x_i^2)^2$. By Cauchy–Schwarz, $(\sum x_i^2)^2 = (\sum x_i^{1/2}x_i^{3/2})^2 \le \sum x_i\sum x_i^3$, so $\operatorname{Var}\hat\beta_{OLS}\ge\sigma^2/\sum x_i$.

(c) $\tilde\sigma^2 = \frac{1}{n-1}\sum(y_i-\hat\beta_G x_i)^2/x_i$.

Solution 41 *When OLS is GLS*

(a) Use $(I+cJ)^{-1} = I-\frac{c}{1+nc}J$ (from $J^2=nJ$) with $c=\rho/(1-\rho)$.

(b) $\mathbf{1}\in\operatorname{col}(X)$, and $V\mathbf{x} = (1-\rho)\mathbf{x}+\rho(\mathbf{1}'\mathbf{x})\mathbf{1}\in\operatorname{col}(X)$ for $\mathbf{x}\in\operatorname{col}(X)$. So $V^{-1}X = XB$ for some invertible B, and $(X'V^{-1}X)^{-1}X'V^{-1} = (B'X'X)^{-1}B'X' = (X'X)^{-1}X'$.

(c) $E\,\mathbf{y}'(I-H)\mathbf{y} = \sigma^2\operatorname{tr}\{(I-H)V\} = \sigma^2(1-\rho)(n-p-1)$, since $(I-H)J=0$. So MSE estimates $(1-\rho)\sigma^2$, not σ^2, and the usual standard error of the intercept is wrong (the slopes' standard errors are correct when computed with $(1-\rho)\sigma^2$).

Solution 42 *GLS is BLUE* **[Past exam 2020-10 #4]**

Unbiasedness gives $AX=I$. Write $A=(X'V^{-1}X)^{-1}X'V^{-1}+D$, so $DX=0$. Then

$$\operatorname{Var}\tilde\beta = AVA' = (X'V^{-1}X)^{-1}+DVD'+(X'V^{-1}X)^{-1}X'D'+DX(X'V^{-1}X)^{-1},$$

and the cross terms vanish because $DX=0$. Hence $\mathbf{a}'(\operatorname{Var}\tilde\beta-\operatorname{Var}\hat\beta)\mathbf{a} = \mathbf{a}'DVD'\mathbf{a} = \|V^{1/2}D'\mathbf{a}\|^2\ge 0$.

Solution 43 *Generalized least squares* **[Past exam 2023-04 #2]**

(1) $E\tilde{\beta} = \beta$ and $\operatorname{Var}\tilde{\beta} = \sigma^2(X'X)^{-1}X'VX(X'X)^{-1}$.

(2) Transform by $V^{-1/2}$ to obtain $\operatorname{Var} = \sigma^2 I$, then apply Gauss–Markov; or see Problem 16.

(3) $\sigma^2(X'V^{-1}X)^{-1}$.

(4) $\tilde{\sigma}^2 = (\mathbf{y} - X\hat{\beta})'V^{-1}(\mathbf{y} - X\hat{\beta})/(n-k-1)$.

(5) Parametrize $V = V(\theta)$ (AR(1), equicorrelation, variance functions), estimate θ from OLS residuals, and plug in (feasible GLS), possibly iterating.

Solution 44 *Variance inflation*

(a) By the partitioned-inverse formula, $c_{jj} = \{\mathbf{x}_j'(I - H_{-j})\mathbf{x}_j\}^{-1}$, the inverse of the residual sum of squares of x_j on the others, which is $1 - R_j^2$ because $\mathbf{x}_j'\mathbf{x}_j = 1$ for a standardized centred column.

(b) $\sum c_{jj} = \operatorname{tr}(X'X)^{-1} = \sum 1/\lambda_j$.

(c) VIF $= 1/(1 - 0.9025) = 10.26$, so standard errors (and interval widths) are $\sqrt{10.26} = 3.2$ times those for uncorrelated regressors.

Solution 45 *Ridge regression*

(a) The gradient of the penalized criterion is $-2X'(\mathbf{y} - X\beta) + 2k\beta$; it vanishes at $\hat{\beta}(k)$, and the Hessian $2(X'X + kI)$ is positive definite. The augmented design gives normal equations $(X'X + kI)\beta = X'\mathbf{y}$.

(b) $E\hat{\beta}(k) = (X'X + kI)^{-1}X'X\beta$ and $\operatorname{Var}\hat{\beta}(k) = \sigma^2(X'X + kI)^{-1}X'X(X'X + kI)^{-1}$.

(c) In canonical coordinates, $g(k) = \sigma^2 \sum \frac{\lambda_j}{(\lambda_j + k)^2} + k^2 \sum \frac{\alpha_j^2}{(\lambda_j + k)^2}$ with $g(0) = E\|\hat{\beta} - \beta\|^2$ and $g'(0) = -2\sigma^2 \sum \lambda_j^{-2} < 0$.

(d) $g(k) = \frac{p\sigma^2 + k^2\|\beta\|^2}{(1+k)^2}$, and $g'(k) = 0$ gives $k = p\sigma^2/\|\beta\|^2$.

Solution 46 *Ridge regression with rank deficiency* **[Past exam 2021-04 #2]**

(a) $\hat{\beta}_{\text{ridge}} = (X'X + kI)^{-1}X'\mathbf{y}$; $X'X + kI$ is positive definite even if X is rank deficient.

(b) $\operatorname{Var} = \sigma^2(X'X + kI)^{-1}X'X(X'X + kI)^{-1} = \sigma^2 P \operatorname{diag}\{\lambda_i/(\lambda_i + k)^2\}P'$ with $X'X = P\Lambda P'$. Take the trace.

Solution 47 *Ridge regression via augmented data* **[Past exam 2021-10 #2]**

$X^{*\prime}X^* = X'X + \lambda I$ and $X^{*\prime}\mathbf{y}^* = X'\mathbf{y}$, so $\hat{\beta}$ is the ridge estimator. With $X = UDV'$,

$$\hat{\beta} = V(D^2 + \lambda I)^{-1}DU'\mathbf{y},$$

$$E(\hat{\beta} \mid X) = V(D^2 + \lambda I)^{-1}D^2V'\beta, \qquad \operatorname{Var}(\hat{\beta} \mid X) = \sigma^2 V(D^2 + \lambda I)^{-2}D^2V'.$$

Solution 48 *Ridge versus least squares* **[Past exam 2025-10 #5]**

(a) $E\hat{\beta}_R = (X'X + \lambda I)^{-1}X'X\beta$ and $\operatorname{Var}\hat{\beta}_R = \sigma^2(X'X + \lambda I)^{-1}X'X(X'X + \lambda I)^{-1}$.

(b) With $X'X = P\Lambda P'$, $\operatorname{Var}\hat{\beta} - \operatorname{Var}\hat{\beta}_R = \sigma^2 P \operatorname{diag}\left\{\frac{1}{\lambda_j} - \frac{\lambda_j}{(\lambda_j+\lambda)^2}\right\}P'$, and each entry is positive for every $\lambda > 0$: the variance is always smaller (and the mean squared error is smaller for small λ).

(c) $\lambda \to 0$: $\hat{\beta}_R \to \hat{\beta}$ (OLS). $\lambda \to \infty$: $\hat{\beta}_R \approx X'\mathbf{y}/\lambda \to \mathbf{0}$ (all coefficients shrunk to zero; zero variance, maximal bias).

Solution 49 *Principal component regression*

(a) $E\tilde{\beta}_r = \sum_{k\le r}\mathbf{e}_k\mathbf{e}_k'\beta$, so the bias is $-\sum_{k>r}\mathbf{e}_k\mathbf{e}_k'\beta$; $\operatorname{Var}\tilde{\beta}_r = \sigma^2\sum_{k\le r} l_k^{-1}\mathbf{e}_k\mathbf{e}_k'$.

(b) $E\|\tilde{\beta}_r - \beta\|^2 = \sigma^2\sum_{k\le r} l_k^{-1} + \sum_{k>r}(\mathbf{e}_k'\beta)^2$, versus $\sigma^2\sum_k l_k^{-1}$ for OLS; subtract.

(c) Canonical coefficient k is multiplied by 1 for $k \le r$ and 0 for $k > r$ in PCR, and by $l_k/(l_k + c)$ in ridge with constant c: PCR is hard thresholding, ridge smooth shrinkage.

Solution 50 *Subgradients and penalized estimation* **[Past exam 2024-10 #5]**

(a) $0 \in \partial L(\theta)$ iff $L(\phi) \ge L(\theta)$ for all ϕ.

(b) For $\theta > 0$, the inequality $|\phi| \ge \theta + z(\phi - \theta)$ for ϕ near θ forces $z = 1$; similarly for $\theta < 0$. At 0, $|\phi| \ge z\phi$ for all ϕ iff $|z| \le 1$.

(c), (d) With $S = \sum x_i^2$ and centred data, $L_\lambda(\theta) = \frac{S}{2}(\theta - \theta_0)^2 + \lambda|\theta| + \text{const}$. So $\partial L_\lambda(\theta) = S(\theta - \theta_0) + \lambda\partial|\theta|$. For $\theta > 0$ we need $\theta = \theta_0 - \lambda/S > 0$; for $\theta < 0$, $\theta = \theta_0 + \lambda/S < 0$; and $\theta = 0$ works iff $|S\theta_0| \le \lambda$. Together this is $\hat{\theta}_\lambda = \operatorname{sgn}(\theta_0)(|\theta_0| - \lambda/S)_+$ (soft thresholding). It is unique because L_λ is strictly convex ($S > 0$).

F. Logistic regression

Solution 51 *Logistic regression: estimation*

(a) $l = \sum\{y_i\mathbf{x}_i'\beta - \log(1 + e^{\mathbf{x}_i'\beta})\}$, so the score is $X'(\mathbf{y} - \pi)$ and the Hessian is $-X'WX$ with $W = \operatorname{diag}\{\pi_i(1 - \pi_i)\}$, which is negative definite.

(b) $\beta^{new} = \beta + (X'WX)^{-1}X'(\mathbf{y} - \pi) = (X'WX)^{-1}X'W\mathbf{z}$ with $\mathbf{z} = X\beta + W^{-1}(\mathbf{y} - \pi)$: weighted least squares of $\mathbf{z}$ on X with weights W.

(c) e^{β_j} is the multiplicative change in the odds $P(y = 1)/P(y = 0)$ when x_j increases by one unit, the others fixed.

Solution 52 *Logistic regression:* 2×2 *tables*

(a) The model is saturated, so the MLEs of $P(y = 1 \mid x)$ are the observed proportions n_{x1}/n_{x+}; then $e^{\hat{\beta}_1}$ is the ratio of the observed odds, $\frac{n_{11}/n_{10}}{n_{01}/n_{00}}$.

(b) Both odds ratios equal $\frac{p_{11}p_{00}}{p_{10}p_{01}}$ in terms of the cell probabilities. In a case–control study only $P(x \mid y)$ is estimable, yet it gives the same odds ratio, so the logistic slope is estimable.

(c) $\widehat{OR} = \frac{20 \cdot 90}{80 \cdot 10} = 2.25$, $\log \widehat{OR} = 0.811$, and se $= \sqrt{\frac{1}{20} + \frac{1}{80} + \frac{1}{10} + \frac{1}{90}} = 0.416$. The interval is $\exp(0.811 \pm 0.815) = (1.00, 5.08)$, borderline.

Solution 53 *Separation*

(a) Along $\beta = t(-2.5, 1)$ the linear predictors are $t(-1.5, -0.5, 0.5, 1.5)$, negative exactly for the zeros. Each term $y_i\eta_i - \log(1 + e^{\eta_i}) = -\log(1 + e^{-|\eta_i|})$ increases to 0 as $t \to \infty$.

(b) The supremum 0 is not attained by any finite β, since every term is < 0. So there is no MLE (complete separation).

(c) No: the least squares criterion is a quadratic with positive definite Hessian (full rank), so a minimizer always exists.